



**Hewlett Packard**  
Enterprise

## **Running Linux on HPE Superdome Flex 280 Server**

### **Abstract**

Instructions to deploy and configure Linux operating systems on HPE Superdome Flex 280 Server

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# Supported operating systems

HPE Superdome Flex 280 Server supports the following operating systems.

**Table 1: OS support for HPE Superdome Flex 280 Server**

| Linux OS                     | Supported versions           |
|------------------------------|------------------------------|
| Red Hat Enterprise Linux     | RHEL 7.9, RHEL 8.2, 8.3      |
| Microsoft Windows            | Windows 2019<br>Windows 2016 |
| VMware                       | VMware vSphere 7.0 U1        |
| SUSE Linux Enterprise Server | SLES 15 SP2                  |

For the specific hardware configurations certified, see the OS vendor distribution sites.

- Red Hat Catalog:  
<https://catalog.redhat.com/hardware/servers/search>
- SLES Catalog:  
<https://www.suse.com/yessearch/>

## HPE Foundation Software

The Linux HPE Foundation Software (HFS) provides updated kernel modules and scripts for use on HPE Foundation Software. It supports Red Hat Linux (RHEL), SUSE Linux Enterprise Server (SLES), and Oracle Linux. HFS consists of software packages designed to ensure the smooth operation of HPE Superdome Flex 280 Server. HFS includes Data Collection Daemon (DCD), an agentless service that proactively monitors the health of hardware components.

## Supported HPE Foundation Software releases

Linux OS environments on HPE Superdome Flex 280 Server require that HPE Foundation Software is also installed.

For information about server hardware supported by each HFS release, see the HPE Foundation Software download page.

Go to [www.hpe.com/support/superdomeflex280-software](http://www.hpe.com/support/superdomeflex280-software) and select `Software` (entitlement required). Select any bundle and click the **Revision History** tab to view information about all HFS versions.

### HFS 2.4

The HPE Foundation Software 2.4 release supports the following Linux operating systems on HPE Superdome Flex 280 Server.

- Red Hat Enterprise Linux 7.9
- Red Hat Enterprise Linux 8.2, 8.3
- SUSE Linux Enterprise Server 15 SP2



# HPE Data Collection Daemon

Data Collection Daemon (DCD) is an agentless service for HPE Superdome Flex Family. DCD proactively monitors the health of hardware components that are visible to the running operating system instance. Any errors are reported to management firmware running on the Rack Management Controller (RMC) for the server. The management service running on the RMC uses the data within its Redfish service and serves it out of band to client applications like HPE Insight Remote Support (IRS).

DCD is bundled with the HPE Foundation Software and includes the following major features for Linux:

- DCD collects inventory data for the following listed components:
  - SAS/SATA controllers
  - Physical drives— Base Chassis Drives (Internal only)
  - RAID volumes (Logical drives)
  - Fibre Channel devices
  - Ethernet devices
  - NVMe controllers
  - GPU cards
  - Linux Host Operating System
- DCD proactively monitors the health of the host OS, SAS/SATA controllers, attached drives, RAID volumes, Fibre Channel devices, Ethernet devices, and NVMe controllers. Any minor or critical state change events are auto-forwarded to the RMC.
- The `dcdcli` command-line utility is provided to test the health of DCD service and its connectivity to the RMC and its clients.
- Logging is supported in DCD to log messages at different logging levels.
- All the data is transferred between DCD and RMC through an in-band IPMI Channel.
- DCD is delivered as an HPE signed and certified package.

## HPE Superdome Flex 280 Server system firmware bundle

The latest firmware recipe bundle for HPE Superdome Flex 280 Server system is available at [www.hpe.com/support/superdomeflex280-software](http://www.hpe.com/support/superdomeflex280-software). Hewlett Packard Enterprise recommends running the latest firmware on your systems.

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 **IMPORTANT:** Do not stop the firmware update while it is in progress. Doing so can place the system in an unusable state. Make sure that AC power to the system remains in place throughout the update process.

---

1. On the landing page, select the appropriate operating system.
2. Choose either the **Manual** method, Smart Update Manager (**SUM**) method, or HPE OneView method from the firmware bundles labeled as **Firmware (Entitlement Required)**.



- **Manual** method: Choose the firmware bundle labeled **for manual installation**. This bundle can be installed or updated directly from the RMC with the installation instructions in the bundle, under *Installation instructions*.
- **SUM** method: Choose the firmware bundle labeled **for SUM installation**. This bundle can be installed or updated from any system on the same network as the RMC using the included copy of the SUM tool. Follow the instructions in the bundle, under *Installation instructions*.
- **HPE OneView** method: Choose the firmware bundle labeled **for OneView installation**. This bundle can be installed from the HPE OneView application. Follow the instructions in the bundle, under *Installation instructions*.

---

**NOTE:** If a firmware mismatch error is displayed after an update, retry the update. If you continue to see failures, contact a support specialist.

---

For a list of Superdome Flex 280 Server components, see [www.hpe.com/support/superdomeflex280-release-sets](http://www.hpe.com/support/superdomeflex280-release-sets).

## HPE Superdome Flex 280 Server I/O Service Pack

Customers can use the HPE Superdome Flex 280 Server I/O Service Pack to update I/O firmware using HPE Smart Update Manager. The HPE Superdome Flex 280 Server I/O Service Pack ISO is available at [www.hpe.com/support/superdomeflex280-software](http://www.hpe.com/support/superdomeflex280-software). This service pack ISO image can be installed/updated with the install instructions in the bundle under installation instructions.

Updated I/O drivers for HPE Superdome Flex 280 Server system are available in the I/O Service Pack. Install the I/O drivers after installing the operating system.



# Logging in to the system

## Logging in to the RMC web GUI

The RMC web GUI provides a management interface accessible by a web browser.

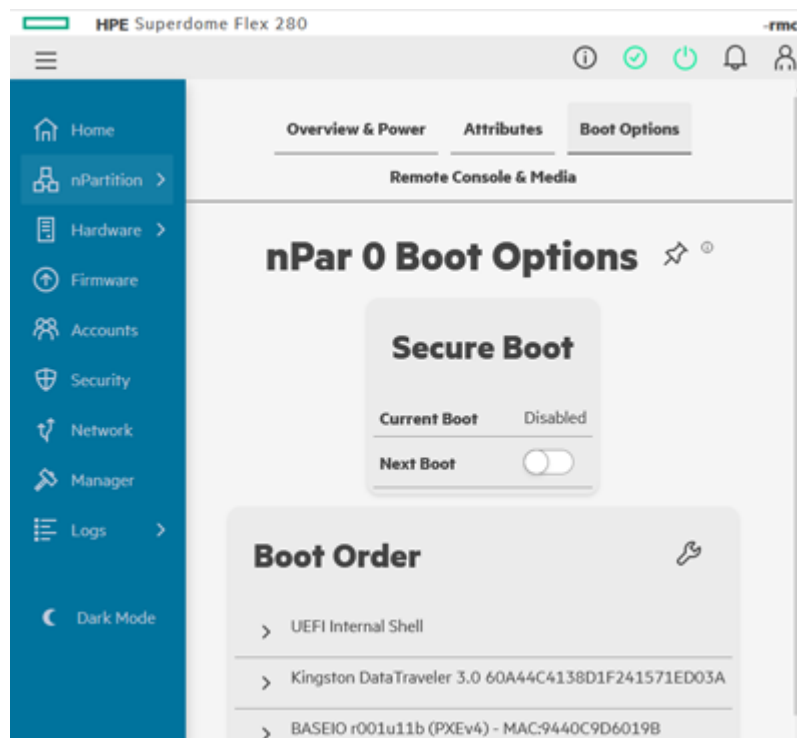
### Prerequisites

- An Ethernet cable connects the base chassis eRMC port to the manageability network.
- An IP address is configured for the eRMC.

### Procedure

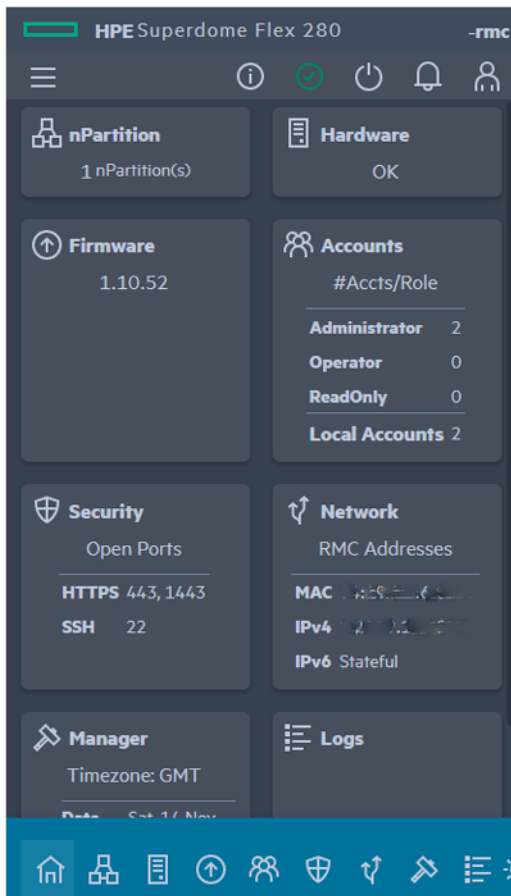
1. Use a web browser to access the RMC web GUI at `https://RMC-IP-ADDRESS`.
2. Log in with an RMC user account and password.

You can operate the RMC web GUI from a phone or desktop browser interface.



**Figure 1: RMC web GUI browser view**





**Figure 2: RMC web GUI phone view**

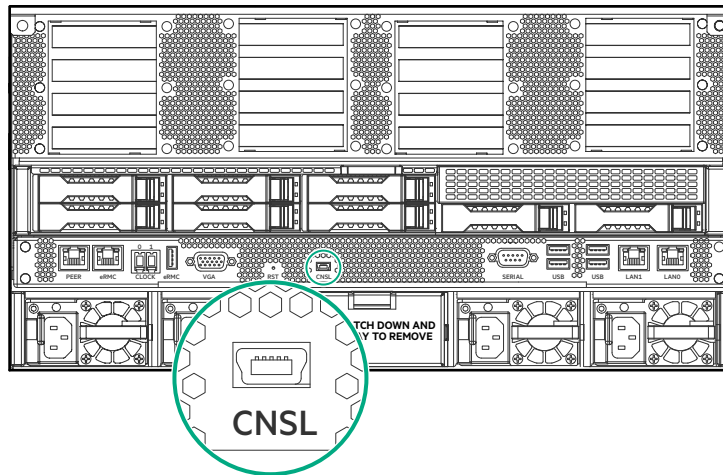
## Logging in to the RMC CLI through the CNSL port (Windows)

### Prerequisites

The FT230X device driver must be installed. You can download the device driver and setup instructions from <https://www.ftdichip.com/Drivers/D2XX.htm>. Windows 10 systems do not have the FT230X driver installed by default.

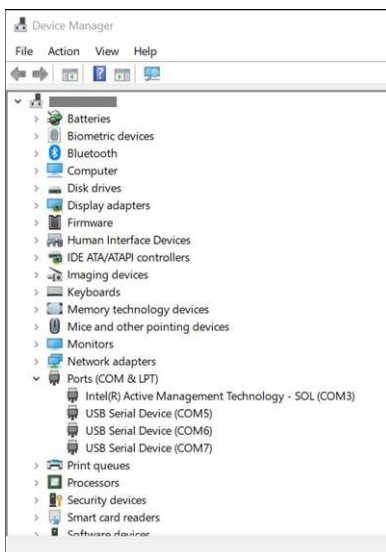
### Procedure

1. Connect a mini-USB cable between the Windows laptop port and the base chassis CNSL port.



**Figure 3: CNSL port on chassis rear**

2. In Windows, use **Settings > Device Manager > Ports (COM & LPT)** to list the available COM ports.



3. Determine which COM port is assigned to the RMC. The BMC port enables RMC CLI access.

In Windows, the Superdome Flex 280 Server port numbering can vary.

Three COM ports represent the CNSL port. One COM port is the BMC port (RMC CLI), one is the SMC port (unused, no customer or service feature), and the other port also is unused.

4. Use PuTTY or another terminal program to connect to the COM port.

Establish a serial connection at 9600 or 115200 baud with 8 data bits, 1 stop bit, no parity, XON/XOFF flow control.

5. Press Enter to access the RMC CLI login prompt.

```
example-rmc login: USER-NAME
Password: PASSWORD
```

```
HPE SD Flex 280 BMC, Firmware Rev. 3.0.xxxxxxx
(C) Copyright 2017-2020 Hewlett Packard Enterprise Development LP
```



```
Type "help" to see list of available commands.  
Type "help <command>" to learn more about each command.
```

```
Enter <tab> to tab-complete a command.  
Use cursor keys for command history.
```

```
HPE Rack Management Controller  
(C) Copyright 2017-2020 Hewlett Packard Enterprise Development LP
```

```
=====
```

```
example-rmc eRMC:r001u01c cli> help
```

```
Commands (type "help <command>" for more information):
```

```
=====
```

|         |          |          |          |        |         |        |        |
|---------|----------|----------|----------|--------|---------|--------|--------|
| acquit  | clear    | deconfig | exit     | ipmi   | power   | save   | show   |
| add     | collect  | disable  | generate | modify | reboot  | search | test   |
| apropos | commands | download | help     | ping   | remove  | set    | update |
| cancel  | connect  | enable   | indict   | ping6  | restore | shell  | upload |

```
example-rmc eRMC:r001u01c cli>
```

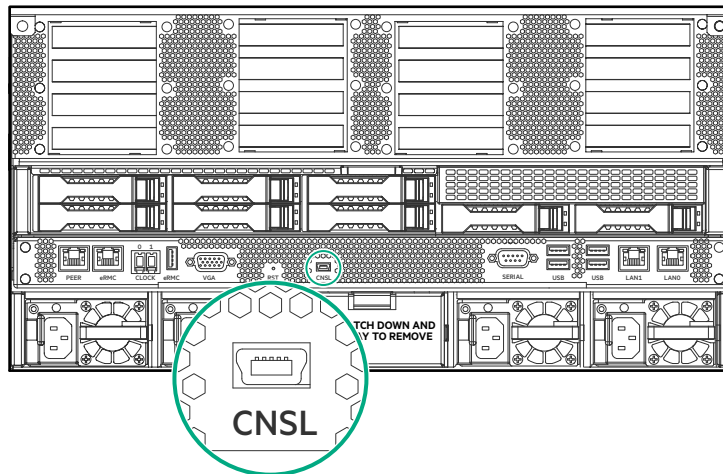
## Logging in to the RMC through the CNSL port (Linux)

### Prerequisites

The FT230X device driver must be installed. You can download the device driver and setup instructions from <https://www.ftdichip.com/Drivers/D2XX.htm>. Linux systems might have the FT230X driver installed by default.

### Procedure

1. Connect a mini-USB cable between the laptop and the CNSL port on the base chassis.



**Figure 4: CNSL port on chassis rear**

2. Use `cu` or `minicom` to connect to the `/dev/ttyACM1` device.

```
linux# minicom -D /dev/ttyACM1
```

Connect at 9600 or 115200 baud using 8 data bits, 1 stop bit, no parity, XON/XOFF flow control.

3. Press Enter to access the RMC CLI prompt.



# Gathering installation materials and information

The following procedure explains how to gather the information you need for the operating system installation session. If you gather the information you need in advance, you can complete the installation more quickly.

## Procedure

1. Obtain the operating system software from the operating system vendor and write this software to a DVD or an appropriately formatted USB storage device.
2. Verify your site credentials, site registration status, and other customer data with your operating system vendor.
3. Obtain the HPE Foundation Software.

Hewlett Packard Enterprise recommends that you read the HFS release notes before you install HFS. The HFS release notes and the software download are at [www.hpe.com/support/superdomeflex280-software](http://www.hpe.com/support/superdomeflex280-software).

Write the ISO file to a DVD, to a USB storage device, or to a location on your local network.

4. Gather the information that the operating system installer requires.

The installation requires you to provide information about passwords, your public (or in-house) network, and so on. If you gather the following information from your site network admin before you begin, you can complete the installation more quickly. Be sure to store this data securely and destroy insecure copies.

- Server fully qualified domain name (FQDN) \_\_\_\_\_
- Server hostname \_\_\_\_\_
- Server IP address \_\_\_\_\_
- Server subnet mask \_\_\_\_\_
- Server administrator user password \_\_\_\_\_
- Site DNS server IP addresses \_\_\_\_\_
- Site search domain \_\_\_\_\_
- FQDN of your site network time protocol (NTP) server \_\_\_\_\_
- Primary name server IP address \_\_\_\_\_
- Secondary name server IP address \_\_\_\_\_
- Rack management controller (RMC) fully qualified domain name (FQDN) \_\_\_\_\_
- RMC administrator user password \_\_\_\_\_

5. Connect to the server to enable installing the OS.



# Installing Red Hat Enterprise Linux

This procedure explains how to install and configure the following software.

- The RHEL operating system.
- The HPE Foundation Software (HFS).

See **Supported HFS version with Linux OS versions** for versions that support your Linux OS.

## Procedure

### Installing the OS and HPE Foundation Software

1. **Initiating RHEL installation and partitioning the disk**
2. **Configuring RHEL network and rebooting**
3. **Completing RHEL installation**
4. **Installing HPE Foundation Software on RHEL**

## Initiating RHEL installation and partitioning the disk

### Prerequisites

Gather installation materials and information you need for the operating system installation.

### Procedure

1. On the boot loader menu, complete the following steps:
  - a. Use the arrow keys to select **Install Red Hat Enterprise Linux x.x**.
  - b. Press **E** (for edit).  
  
Edit the installer kernel boot line to remove `quiet` and add `erst_disable edac_report=off console=ttyS0,115200 earlyprintk-ttyS0,115200 bau=0 mce=2 nmi_watchdog=0 pci=nobar`
  - c. Press **Ctrl-X** to start the boot.
2. Wait a few moments while the software loads.
3. On the **WELCOME ...** screen, which asks **What language would you like to use during the installation process?**, complete the following steps:
  - a. Select your language.
  - b. Click **Continue**.
4. On the **INSTALLATION SUMMARY** page, click **DATE & TIME**.
5. On the **DATE & TIME** page, complete the following steps:



- a. Select your time zone.
  - b. Select the date.
  - c. Click **Done**.
6. On the **INSTALLATION SUMMARY** page, click **KEYBOARD**.
7. On the **KEYBOARD LAYOUT** page, complete the following steps:
  - a. Select your keyboard layout.
  - b. Click **Done**.
8. On the **INSTALLATION SUMMARY** page, click **SOFTWARE SELECTION**.
9. On the **SOFTWARE SELECTION** page, complete the following steps:
  - a. Select **Server with GUI**.
  - b. Click **Done**.
10. On the **INSTALLATION SUMMARY** page, complete the following steps:
  - a. Click **INSTALLATION DESTINATION**.
  - b. Under **Local Standard Disks**, click the disk onto which you want to install the operating system.
  - c. Under **Other Storage Options**, click **I will configure partitioning**.
  - d. Click **Done**.
11. On the **MANUAL PARTITIONING** page, clean the disk.

The screen left pane lists the operating system installations that currently reside on the disk. Your goal is to remove all operating system installations, data, and partitions that reside on the disk. You can remove one operating system at a time.

To remove one operating system, complete the following steps:

  - a. Select the operating system name.
  - b. Click the minus sign (-) at the bottom of the left pane to delete the operating system.
  - c. On the **Are you sure ...** popup, complete the following steps:
    - I. Select **Delete all other ...**.
    - II. Click **Delete it**.
  - d. Repeat the preceding steps, as needed, until all operating systems are removed.
12. In the left pane, from the **New mount points will use the following partition scheme** drop-down list, select **Standard Partition**.
13. Create mount points.

Configure the disks according to the following table.



| Size                | Mount point | Filesystem |
|---------------------|-------------|------------|
| 512 MB              | /boot/efi   | FAT16      |
| 250 GB (256,000 MB) | /           | XFS        |
| 8 GB (8,192 MB)     | swap        | swap       |
| rest of disk        | /data1      | XFS        |

To create mount points, complete the following steps:

- a. In the left pane, click the plus sign (+) to add a mount point.
- b. On the **ADD A NEW MOUNT POINT** popup, complete the following steps:
  - I. On the **Mount Point** drop-down menu, select **/boot/efi**.
  - II. On the **Desired Capacity** field, enter 512mb.
  - III. Click **Add mount point**.
- c. In the left pane, click the plus sign (+) to add a mount point.
- d. On the **ADD A NEW MOUNT POINT** popup, complete the following steps:
  - I. On the **Mount Point** drop-down menu, select **/**.
  - II. On the **Desired Capacity** field, enter 250gb.
  - III. Click **Add mount point**.
- e. In the left pane, click the plus sign (+) to add a mount point.
- f. On the **ADD A NEW MOUNT POINT** popup, complete the following steps:
  - I. On the **Mount Point** drop-down menu, select **swap**.
  - II. On the **Desired Capacity** field, enter 8gb.
  - III. Click **Add mount point**.
- g. In the left pane, click the plus sign (+) to add a mount point.
- h. On the **ADD A NEW MOUNT POINT** popup, complete the following steps:
  - I. In the **Mount Point** field, enter /data1.
  - II. Leave the **Desired Capacity** field blank.
  - III. Click **Add mount point**.
- i. Verify that the **Desired Capacity** field shows the rest of the disk.
- j. Click **Done**.





14. On the **SUMMARY OF CHANGES** popup, click **Accept Changes**.
15. On the **INSTALLATION SUMMARY** page, click **KDUMP**.
16. On the **KDUMP** page, complete the following steps:
  - a. Next to **Kdump Memory Reservation**, click **Manual**.
  - b. In the **Memory to be Reserved (MB)** field, specify 450 MB.
  - c. Click **Done**.

More information

[Installing Red Hat Enterprise Linux](#)  
[Configuring RHEL network and rebooting](#)

Configuring RHEL network and rebooting

Procedure

1. On the **INSTALLATION SUMMARY** page, click **NETWORK & HOST NAME**.
2. On the **NETWORK & HOST NAME** page, complete the following steps:
  - a. Select the Ethernet device that you want to configure.
  - b. In the right pane, in the upper-right corner of the screen, find the **ON/OFF** switch, and click the blank box to set the **ON/OFF** switch to **ON**.
  - c. Click **Configure**.
  - d. On the **Editing device\_name** popup, select **IPv4 Settings**.
  - e. Complete this step according to the following table:

| To specify dynamic addressing:                                            | To specify static addressing:                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. In the <b>Method:</b> drop-down menu, select <b>Automatic (DHCP)</b> . | 1. In the <b>Method:</b> drop-down menu, select <b>Manual</b> .                                                                                                                                                                                 |
| 2. On the <b>Editing</b> pop-up, click <b>Save</b> .                      | 2. In the <b>Addresses</b> pane, click <b>Add</b> , and complete the following steps: <ul style="list-style-type: none"> <li>Enter the IP address.</li> <li>Enter the netmask.</li> <li>Enter the IP address of the default gateway.</li> </ul> |
|                                                                           | 3. In the <b>DNS servers:</b> field, enter the IP address of one or more DNS servers. If you specify more than one, use a comma to separate each IP address.                                                                                    |

Table Continued



**To specify dynamic addressing:****To specify static addressing:**

4. In the **Search domains:** field, enter one or more search domains. If you specify more than one, use a comma to separate each domain.

5. On the **Editing** pop-up, click **Save**.

f. In the **Host name** field, enter the hostname of the system.

g. Click **Done**.

3. On the **INSTALLATION SUMMARY** page, click **Begin Installation**.

4. On the **CONFIGURATION** page, click **ROOT PASSWORD**.

5. On the **ROOT PASSWORD** page, complete the following steps:

a. In the **Root Password** field, enter the password you want to use on this system.

b. In the **Confirm** field, enter the password again.

c. Click **Done**.

If the password is too weak, either specify a stronger password or click **Done** twice.

6. On the **CONFIGURATION** page, complete the following steps:

a. Wait for the installation to complete.

b. Click **Reboot** when the installation completes.

c. Wait for the reboot to finish.

The OS auto boots after being rebooted. Wait for the GRUB menu and complete the following step.

d. When the GRUB menu appears, press **e** to edit the top command line.

The top line is highlighted to select the RHEL version. Do **NOT** select the rescue mode version.

7. To reset the boot parameters, complete the following steps:

a. On the boot loader menu, use the arrow keys to select **Install Red Hat Enterprise Linux x.x**.

b. Press **E** (for edit).

```
Edit the installer kernel boot line to remove quiet and add
erst_disable edac_report=off
console=ttyS0,115200 earlyprintk=ttyS0,115200 bau=0 mce=2 nmi_watchdog=0
pci=nobar
```

c. To start the boot, press **Ctrl-X**.

d. Enter the following string at the end of the `linuxefi` kernel command line:

```
modprobe.blacklist=skx_edac
```

Use a backslash (`\`) character if needed, to continue the line appropriately. The following figure shows the newly edited file and the use of the backslash.



```

insmod part_gpt
insmod xfs
set root='hd0,gpt2'
if [ x$feature_platform_search_hint = xy ]; then
    search --no-floppy --fs-uuid --set=root --hint-bios=hd0,gpt2 --hint-\
efi=hd0,gpt2 --hint-baremetal=ahci0,gpt2 01ff50f3-90cf-426d-a2f1-083508c0d1bc
else
    search --no-floppy --fs-uuid --set=root 01ff50f3-90cf-426d-a2f1-0835\
08c0d1bc
fi
linuxefi /boot/vmlinuz-3.10.0-693.el7.x86_64 root=UUID=01ff50f3-90cf-4\
26d-a2f1-083508c0d1bc ro crashkernel=450M rhgb quiet LANG=en_US.UTF-8 modprobe\
.blacklist=skx_edac_
initrdefi /boot/initramfs-3.10.0-693.el7.x86_64.img

```

Press Ctrl-x to start, Ctrl-c for a command prompt or Escape to discard edits and return to the menu. Pressing Tab lists possible completions.

e. Press **CTRL-X** to start the boot.

8. (Optional) On the **CONFIGURATION** page, click **USER CREATION** and follow the prompts.

Complete this optional step if you want to configure additional user accounts.

#### More information

[Installing Red Hat Enterprise Linux](#)

[Completing RHEL installation](#)

## Completing RHEL installation

**NOTE:** If a read/write console connection from the RMC is active, Steps 1-5 may appear on the active console connection rather than on the local attached keyboard, monitor, and mouse.

#### Procedure

1. On the **INITIAL SETUP** page, click **LICENSE INFORMATION**.
2. On the **LICENSE INFORMATION** page, complete the following steps:
  - a. Click **I accept the license agreement**.
  - b. Click **Done**.
3. On the **INITIAL SETUP** page, click **Subscription Manager**.
4. On the **Subscription Manager** page, complete the following steps:

- a. Complete the fields according to your site practices.
  - b. Click **Done**.
5. Click **FINISH CONFIGURATION**.
6. On the **Welcome** page, complete the following steps:
  - a. Select your language.
  - b. Click **Next**.
7. On the **Typing** page, complete the following steps:
  - a. Select the language you use.
  - b. Click **Next**.
8. On the **Privacy** page, complete the following steps:
  - a. In the right pane, in the upper-right corner of the screen, find the **ON/OFF** switch. Click the blank box to set **Location Services** to **OFF**.
  - b. Click **Next**.
9. On the **Time Zone** page, complete the following steps:
  - a. Type your location, click the magnifying glass search icon, and press **Enter**.
  - b. Click **Next**.
10. On the **Connect Your Online Accounts** page, click **Skip**.
11. On the **About You** page, complete all fields and click **Next**.
12. On the **Set a Password** page, complete all fields and click **Next**.
13. On the **You're ready to go** page, click **Start using Red Hat Enterprise Linux Server**.

#### More information

[Installing Red Hat Enterprise Linux](#)

[Installing HPE Foundation Software on RHEL](#)

## Installing HPE Foundation Software on RHEL

To install the HPE Foundation Software on RHEL platforms, complete the following procedure.

---

**NOTE:** The installation of HPE Foundation Software requires additional RHEL packages that were not installed during the initial installation of "Server with GUI" packages. Configure an OS installation repository prior to configuring an HFS repository.

---

#### Procedure

1. Insert the *HPE Foundation Software X.X* media into the DVD drive or navigate to the HFS location on the network.
2. Open a terminal window to the server.



For example, click **Applications > System Tools > Terminal**.

3. Make sure that you are logged in as the root user.

4. To create an installation directory for the files from the media, enter the following command:

```
# mkdir -p /opt/hpe/Factory-Install/hpe-foundation-X.X/
```

5. To mount the media in read-only mode (`-r`), using a loop device, enter the following command:

```
# mount -t iso9660 -ro loop /dev/cdrom /mnt
```

6. To copy the files from the temporary mount directory to the installation directory, enter the following command:

```
# rsync -avHx /mnt/ /opt/hpe/Factory-Install/hpe-foundation-X.X/
```

7. To unmount the media from the temporary directory, enter the following command:

```
# umount /mnt
```

8. Use a text editor to create file `/etc/yum.repos.d/foundation X.X-local.repo` with the following contents:

```
[foundationX.X-repo]
name=HPE Foundation Software X.X - $basearch
baseurl=file:///opt/hpe/Factory-Install/hpe-foundation-X.X/RPMS
enabled=1
gpgcheck=0
gpgkey=file:///opt/hpe/Factory-Install/hpe-foundation-X.X/RPM-GPG-KEY-hpe
      file:///opt/hpe/Factory-Install/hpe-foundation-X.X/RPM-GPG-KEY-sgi
```

9. To display the list of software that you can install and verify that the list includes the foundation software, enter the following command.

```
# yum grouplist | grep HPE
```

10. To install the HPE Foundation Software group, enter the following command:

```
# yum groupinstall "HPE Foundation Software"
```

11. To confirm the download size, enter **y** at the following prompt:

```
Is this ok [y/N]
```

12. To accept the license keys, enter **y** at the following prompt:

```
Is this ok [y/N]
```

The system might prompt you accept one or more `RPM-GPG-KEY-hpe` or `RPM-GPG-KEY-sgi` license key. Accept all the keys.

13. Open a terminal window on the booted server.

14. To reboot the server, enter the following command in the terminal window:

```
# reboot
```



# Installing SUSE Linux Enterprise Server

This procedure explains how to install and configure the following software:

- The SLES operating system.
- The HPE Foundation Software.

## Procedure

### Installing the OS and HPE Foundation Software

1. Initiating SLES installation
2. Using SLES to partition the disk
3. Configuring SLES network and miscellaneous settings
4. Installing HPE Foundation Software on SLES platforms

## Initiating SLES installation

### Prerequisites

- Gather installation materials and information you need for the operating system installation.
- SLES 15 OS installation requires two DVD or ISO images. Use the installation procedure in **OS installation using multiple ISO files** to install SLES 15.

### Procedure

1. On the **SUSE Linux Enterprise Server x** screen, which is the boot loader screen, select **Installation**.
  - a. On the grub menu, select **Advanced options for SLES x**.  
Press E (for edit).
  - b. Edit the boot line, adding: `edac_report=off console=ttyS0,115200 earlyprintk=ttyS0,115200 bau=0 mce=2 nmi_watchdog=0 pci=nobar`
  - c. Press **Ctrl-X** to start the boot.
2. Wait a few moments while the software loads.
3. On the **Language, Keyboard and License Agreement** page, complete the following steps:
  - a. Use the pull-down menu to select your language.
  - b. Use the pull-down menu to select your keyboard layout.
  - c. Check **I Agree to the License Terms**.
  - d. Click **Next**.
4. On the **System Probing ...** screen, monitor the progress.



5. On the **Registration** screen, complete the following steps:
  - a. Provide your site credentials.
  - b. Click **Next**.
6. On the **Add On Product** screen, click **Next**.
7. On the **System Role** screen, select **Default System**, and click **Next**.
8. On the **Suggested Partitioning** screen, click **Expert Partitioner**.

#### More information

[Using SLES to partition the disk](#)  
[Installing SUSE Linux Enterprise Server](#)

## Using SLES to partition the disk

This topic explains how to partition the disk and how to monitor a reboot.

Configure the disks according to the following table.

**Table 2: Disk partitions**

| Size                | Mount Point | Filesystem |
|---------------------|-------------|------------|
| 512 MB              | /boot/efi   | FAT16      |
| 250 GB (256,000 MB) | /           | XFS        |
| 8 GB (8,192 MB)     | swap        | Swap       |
| rest of disk        | /data1      | XFS        |

#### Procedure

1. On the **Expert Partitioner** screen, complete the following steps to clean the disk:
  - a. Expand **Hard Disks**.
  - b. Select the disk you want to use.  
For example, select **sda**.
  - c. On the **Expert ...** drop-down in the lower right part of the screen, select **Create New Partition Table**.
  - d. On the YaST2 pop-up with the message **Really create a new partition table ...**, click **Yes**.
  - e. Expand **Hard Disks**.
  - f. Select the hard disks you want to use.
  - g. Click **Add**.
2. Complete the following steps on the **Add Partition on /dev/disk** screen to configure the boot partition:



- a. Under **New Partition Size**, click **Custom Size**.
  - b. Enter 512 MB in the **Size** field.
  - c. Click **Next**.
  - d. Under the **Role** list, complete the following steps:
    - I. Select **Operating System**.
    - II. Click **Next**.
  - e. Under **Formatting Options**, select **Format partition**.
  - f. Under **File system**, use the pull-down menu to select **FAT**.
  - g. Under **Mounting Options**, select **Mount Partition**.
  - h. Under **Mount point**, use the pull-down menu to select `/boot/efi`.
  - i. Select **Finish**.
3. On the **Expert Partitioner** screen, click **Add**.
4. Complete the following steps on the **Add Partition on /dev/disk** screen to configure the root partition:
  - a. Under **New Partition Size**, click **Custom Size**.
  - b. Enter 250 GB in the **Size** field.
  - c. Click **Next**.
  - d. Under the **Role** list, complete the following steps:
    - Select **Operating System**.
    - Click **Next**.
  - e. Under **Formatting Options**, select **Format partition**.
  - f. Under **File system**, use the pull-down menu to select **XFS**.
  - g. Under **Mounting Options**, select **Mount Partition**.
  - h. Under **Mount point**, use the pull-down menu to select `/`.
  - i. Select **Finish**.
5. On the **Expert Partitioner** screen, click **Add**.
6. Complete the following steps on the **Add Partition on /dev/disk** screen to configure the swap partition:
  - a. Under **New Partition Size**, click **Custom Size**.
  - b. Specify 8 GB in the **Size** field.
  - c. Click **Next**.
  - d. Under the **Role** list, complete the following steps:





- Select **Swap**.
  - Click **Next**.
- e. Under **Formatting Options**, select **Format partition**.
  - f. Under **File system**, use the pull-down menu to select **Swap**.
  - g. Under **Mounting Options**, select **Mount Partition**.
  - h. Under **Mount point**, use the pull-down menu to select **swap**.
  - i. Select **Finish**.
7. On the **Expert Partitioner** screen, click **Add**.
  8. Complete the following steps on the **Add Partition on /dev/disk** screen to configure the data partition:
    - a. Under **New Partition Size**, click **Maximum Size**.
    - b. Click **Next**.
    - c. Under the **Role** list, complete the following steps:
      - I. Select **Data and ISV Applications**.
      - II. Click **Next**.
    - d. Under **Formatting Options**, select **Format partition**.
    - e. Under **File system**, use the pull-down menu to select **XFS**.
    - f. Under **Mounting Options**, select **Mount Partition**.
    - g. Under **Mount point**, enter **/data1**.
    - h. Select **Finish**.
  9. On the **Expert Partitioner** screen, in the **Partitions** tab, examine the disk partitions.  
 If the partitions match the disk partitions in **Table 2: Disk partitions**, click **Accept**.  
 If the partitions are incorrect, correct the partition specifications.
  10. On the **Suggested Partitioning** screen, click **Next**.
  11. On the **Clock and Time Zone** screen, complete the following steps:
    - a. Select your region.
    - b. Select your time zone.
    - c. Check the box next to **Hardware Clock Set To UTC**.
    - d. Click **Next**.
  12. On the **Local Users** screen, click **Create New User**.  
 Complete the following steps:
    - a. Complete the following fields:



- **User's Full Name**
  - **Username**
  - **Password**
  - **Confirm Password**
- b. Click **Next**.
- 13.** On the **Password for the System Administrator “root”** screen, complete the following steps:
- a. In the **Password for root User** field, enter the root user password.
  - b. In the **Confirm password** field, re-enter the root user password.
  - c. Enter a few characters in the **Test Keyboard Layout** field.  
For example, if you specified a language that includes non-English characters and you include these characters in passwords, enter these characters into this field. This field is a plain text field. You assure yourself that the operating system can recognize these characters when you enter them.
  - d. Click **Next**.
- 14.** On the **Installation Settings** screen, configure additional features as needed and click **Install**.
- 15.** On the **Confirm Installation** pop-up, click **Install**.
- 16.** Monitor the installation and, if you are using hard media, be prepared to remove the installation software media before the boot.
- The installation itself can take several minutes. At the end of the installation, the system boots. The installation software notifies you of this boot. If you can, remove the installation DVD before the system boots.
- If you fail to remove the installation DVD before the final boot, the machine boots from the DVD. In this case, complete the following steps:
- a. Remove the DVD after the boot.
  - b. Press **CTRL-ALT-DEL** to boot the machine again.
  - c. Allow the machine to boot from the hard disk to finish the installation.
- If you are installing over the network, the JViewer software manages the disk.

#### More information

[Configuring SLES network and miscellaneous settings](#)  
[Installing SUSE Linux Enterprise Server](#)

## Configuring SLES network and miscellaneous settings

### Procedure

1. Log into the server as the root user.
2. Click **Applications > System Tools > YaST**.
3. On the **Administrator Settings** screen, click **Network Settings**.



4. On the **YaST2 -- Network Settings** screen, highlight the network card you want to configure, and click **Edit**.
5. On the **Network Card Setup** screen, specify dynamic or static addressing, as follows:

| To specify dynamic addressing:                          | To specify static addressing:                                                                                                                                                                                                                                                   |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Select the type of dynamic addressing that you want. | 1. Verify that <b>eth0</b> appears in the <b>Configuration Name</b> field, and click <b>Statically assigned IP Address</b> .                                                                                                                                                    |
| 2. Click <b>Next</b> to accept the default of DHCP.     | 2. Configure the first NIC ( <b>eth0</b> ) for your house (public) network.<br><br>On the <b>Address</b> tab, specify the following: <ul style="list-style-type: none"> <li>• The <b>IP Address</b></li> <li>• The <b>Subnet Mask</b></li> <li>• The <b>Hostname</b></li> </ul> |
|                                                         | 3. Click <b>Next</b> .                                                                                                                                                                                                                                                          |

6. On the **Network Settings** screen, click the **Hostname/DNS** tab and complete the following steps:

- a. Enter the hostname.
- b. Enter the domain name.
- c. Verify that **Change Hostname via DHCP** is set correctly.

To assign a static IP address, clear the check box to the left of the **Change Hostname via DHCP** label. A later step saves the hostname to the `/etc/hosts` file. Consult your network administrator if you have questions regarding the use of DHCP.

**NOTE:** This procedure explains how to configure a static address on a network card. If you want a different network configuration, for example if you want to configure DHCP, ensure that the check box in this step is checked before you click **Next**. You can use this procedure as a guide and consult the SLES documentation for more specific steps.

- d. The IP address for **Name Server 1**.
- e. (Optional) The IP address for **Name Server 2**.
- f. (Optional) The IP address for **Name Server 3**.
- g. (Optional) In the **Domain Search** field, add additional domains.
- h. Click the **Routing** tab.
- i. On the **Routing** tab, enter the **Default Gateway**, and click **OK**.

#### More information

[Installing HPE Foundation Software on SLES platforms](#)

[Installing SUSE Linux Enterprise Server](#)

# Installing HPE Foundation Software on SLES platforms

## Procedure

1. Insert the *HPE Foundation Software X.X* media into the DVD drive or make a network connection to the HPE Foundation Software repository.
2. Log into the server as the root user.
3. To start the YaST interface, click **Applications > System Tools > YaST**.
4. Under **Software**, click **Software Repositories** to start the SLES repository manager.
5. On the **Configured Software Repositories** screen, click **Add**.
6. On the **Add On Product** screen, select **DVD**, and click **Next**.
7. On the **YaST** pop-up, complete the following steps:
  - a. Select the correct DVD.
  - b. Click **Continue**.
8. On the **Import Untrusted GnuPG Key**, follow this procedure to import a trusted key.

---

**NOTE:** Perform this step if you are going through this process for the first time and you must confirm the security of the Hewlett Packard Enterprise digital key.

---

- a. Download the keys.

Copy the compressed tar file (`HPE-GPG-Public-Keys.tar.gz`) from this link to your local directory and extract the public keys.

**<https://downloads.hpe.com/pub/keys/HPE-GPG-Public-Keys.tar.gz>**

- b. Import the keys for GPG.

For each key that you have unzipped, install the public key using the `gpg --import` command.

```
# gpg --import /path_to_the_key/file_name_of_the_key
```

For example: `gpg --import /path_to_the_key/B1275EA3.pub`

- c. Verify using GPG.

Use the `gpg --verify` command to validate and verify the digital signature of the signed file. The output from the command indicates the validity of the signature. Specify the `.sig` (detached signature) file and the corresponding input file in the command.

```
# gpg --verify filename.sig filename
```

If the level of trust on the key has not been set, you will see a trust level warning similar to the following:

```
gpg: WARNING: This key is not certified with a trusted signature!
gpg: There is no indication that the signature belongs to the owner.
```

Because you have downloaded the key from an SSL secured site by Hewlett Packard Enterprise Company, you can ultimately trust that this public key is indeed from Hewlett Packard Enterprise Company. Therefore edit the key to set the trust level of the key for proper verification.

- d. Find the "key\_name" of the key.

Type the following command and select the key that you must trust:

```
# gpg --list-keys
```

Example of a "key\_name": "Hewlett Packard Enterprise Company RSA 2048 1"

- e. Edit the key.

```
# gpg --edit-key "key_name"
```

Type the command **trust** and select "5" for trusting the key ultimately.

- f. Confirm and enter **quit** to exit.

In the future, you will not see the warning about an untrusted identity when verifying the signature. Example verification:

```
# gpg --verify test.bin.sig test.bin
gpg: Signature made Thu 03 Jan 2013 04:48:47 PM UTC using RSA key ID
5CE2D476
gpg: Good signature from "Hewlett Packard Enterprise Company RSA 2048 1"
```

9. On the **Configured Software Repositories** screen, click **OK**.
10. On the **YaST Control Center** screen, click **Software Management**.
11. Select **View > Patterns**.
12. Scroll down to **HPE Foundation**.
13. Check the box to the left of **HPE Foundation Libraries, Software, and Drivers**, and click **Accept**.
14. On the **Changed Packages** pop-up, click **Continue**.
15. Insert the media into the DVD drive as directed by the prompts on the **Perform Installation** screen.  
If necessary, click **Eject** on the YaST2 pop-up to open the DVD drive. In the **YaST** pop-up, you might need to click **Retry** more than once to read a new media.
16. On the **Installation Report** screen, click **Finish**.
17. Close the YaST session.
18. Open a terminal window on the booted system.
19. Log into the server as the root user.
20. Configure items in `/etc/sysconfig/hpe-auto-config`.
21. In the terminal window, enter the following command to reboot the system.

```
# reboot
```



# Updating HPE Foundation Software using online Software Delivery Repository

HPE Foundation Software is available from the HPE Software Delivery Repository at [\*\*https://downloads.linux.hpe.com/SDR/project/hpe-foundation/\*\*](https://downloads.linux.hpe.com/SDR/project/hpe-foundation/).

The SDR provides a way to install software packages on a Linux system by using `yum` or `zypper`.

## Prerequisites

- An active warranty or support contract is required to access HPE Foundation Software product updates.
- The email associated with the HPE Passport account must be supplied as the username.
- A user-generated token must be supplied as the HTTP password to access the repository.

## Procedure

1. Go to the HPE Foundation Software page.

[\*\*https://downloads.linux.hpe.com/SDR/project/hpe-foundation/\*\*](https://downloads.linux.hpe.com/SDR/project/hpe-foundation/)

2. Generate your user token following the instructions on the site.

3. Access the HPE Foundation Software repository with a web browser, `yum`, or `zypper`.

Use the following URL syntax: `https://EMAIL:TOKEN@update1.linux.hpe.com/repo/hpe-foundation`

4. The first time accessing the SDR from a system, you may subscribe the system by following the instructions on the SDR site.
5. For specific `yum` and `zypper` command options for package installation, see the examples at [\*\*https://downloads.linux.hpe.com/SDR/project/hpe-foundation/\*\*](https://downloads.linux.hpe.com/SDR/project/hpe-foundation/).



# Updating HPE Foundation Software on RHEL

To update the HPE Foundation Software on RHEL platforms, complete the following procedure. Some steps contain platform-specific notes.

**NOTE:** The installation of HPE Foundation Software requires additional RHEL packages that were not installed during the initial installation of "Server with GUI" packages. Configure an OS installation repository prior to configuring an HFS repository.

## Procedure

1. Insert the *HPE Foundation Software X.X* media into the DVD drive or navigate to the HFS location on the network.
2. Open a terminal window to the server.  
For example, click **Applications > System Tools > Terminal**.
3. Make sure that you are logged in as the root user.
4. To create an installation directory for the files from the media, enter the following command:

```
# mkdir -p /opt/hpe/Factory-Install/hpe-foundation-X.X/
```

5. To mount the media in read-only mode (`-r`), using a loop device, enter the following command:

```
# mount -t iso9660 -ro loop /dev/cdrom /mnt
```

6. To copy the files from the temporary mount directory to the installation directory, enter the following command:

```
# rsync -avHx /mnt/ /opt/hpe/Factory-Install/hpe-foundation-X.X/
```

7. To unmount the media from the temporary directory, enter the following command:

```
# umount /mnt
```

8. Use a text editor to create file `/etc/yum.repos.d/foundation X.X-local.repo` with the following contents:

```
[foundationX.X-repo]
name=HPE Foundation Software X.X - $basearch
baseurl=file:///opt/hpe/Factory-Install/hpe-foundation-X.X/RPMS
enabled=1
gpgcheck=0
gpgkey=file:///opt/hpe/Factory-Install/hpe-foundation-X.X/RPM-GPG-KEY-hpe
        file:///opt/hpe/Factory-Install/hpe-foundation-X.X/RPM-GPG-KEY-sgi
```

9. To display the list of software that you can install and verify that the list includes the foundation software, enter the following command.

```
# yum grouplist | grep HPE
```

10. To install the HPE Foundation Software group, enter the following command:

```
# yum update "HPE Foundation Software"
```

11. To confirm the download size, enter **y** at the following prompt:

```
Is this ok [y/N]
```

- 12.** To accept the license keys, enter **y** at the following prompt:

```
Is this ok [y/N]
```

The system might prompt you accept one or more RPM-GPG-KEY-hpe or RPM-GPG-KEY-sgi license key. Accept all the keys.

- 13.** Run `hpe-auto-config` to set the bootline options and other settings.
- 14.** To reboot the server, enter the following command in the terminal window:

```
# reboot
```





# Updating HPE Foundation Software on SLES platforms

To update the HPE Foundation Software on SLES platforms, complete the following procedure.

## Procedure

1. Insert the *HPE Foundation Software X.X* media into the DVD drive or make a network connection to the HPE Foundation Software repository.
2. Log into the server as the root user.
3. To start the YaST interface, click **Applications > System Tools > YaST**.
4. Under **Software**, click **Software Repositories** to start the SLES repository manager.
5. On the **Configured Software Repositories** screen, click **Add**.
6. On the **Add On Product** screen, select **DVD**, and click **Next**.
7. On the **YaST** pop-up, complete the following steps:
  - a. Select the correct DVD.
  - b. Click **Continue**.
8. On the **Import Untrusted GnuPG Key**, follow this procedure to import a trusted key.

---

**NOTE:** Perform this step if you are going through this process for the first time and you must confirm the security of the Hewlett Packard Enterprise digital key.

---

- a. Download the keys.

Copy the compressed tar file (`HPE-GPG-Public-Keys.tar.gz`) from this link to your local directory and extract the public keys.

**<https://downloads.hpe.com/pub/keys/HPE-GPG-Public-Keys.tar.gz>**

- b. Import the keys for GPG.

For each key that you have unzipped, install the public key using the `gpg --import` command.

```
# gpg --import /path_to_the_key/file_name_of_the_key
```

For example: `gpg --import /path_to_the_key/B1275EA3.pub`

- c. Verify using GPG.

Use the `gpg --verify` command to validate and verify the digital signature of the signed file. The output from the command indicates the validity of the signature. Specify the `.sig` (detached signature) file and the corresponding input file in the command.

```
# gpg --verify filename.sig filename
```

If the level of trust on the key has not been set, you will see a trust level warning similar to the following:



```
gpg: WARNING: This key is not certified with a trusted signature!  
gpg: There is no indication that the signature belongs to the owner.
```

Because you have downloaded the key from an SSL secured site by Hewlett Packard Enterprise Company, you can ultimately trust that this public key is indeed from Hewlett Packard Enterprise Company. Therefore edit the key to set the trust level of the key for proper verification.

- d. Find the "key\_name" of the key.

Type the following command and select the key that you must trust:

```
# gpg --list-keys
```

Example of a "key\_name": "Hewlett Packard Enterprise Company RSA 2048 1"

- e. Edit the key.

```
# gpg --edit-key "key_name"
```

Type the command "trust" and select "5" for trusting the key ultimately.

- f. Confirm and enter **quit** to exit.

In the future, you will not see the warning about an untrusted identity when verifying the signature. Example verification:

```
# gpg --verify test.bin.sig test.bin  
gpg: Signature made Thu 03 Jan 2013 04:48:47 PM UTC using RSA key ID  
5CE2D476  
gpg: Good signature from "Hewlett Packard Enterprise Company RSA 2048 1"
```

9. On the **Configured Software Repositories** screen, click **OK**.
10. On the **YaST Control Center** screen, click **Software Management**.
11. Select **View > Patterns**.
12. Scroll down to **HPE Foundation**.
13. Check the box to the left of **HPE Foundation Libraries, Software, and Drivers**, and click **Accept**.
14. On the **Changed Packages** pop-up, click **Continue**.
15. Insert the media into the DVD drive as directed by the prompts on the **Perform Installation** screen.  
If necessary, click **Eject** on the YaST2 pop-up to open the DVD drive. In the **YaST** pop-up, you might need to click **Retry** more than once to read a new media.
16. On the **Installation Report** screen, click **Finish**.
17. Close the YaST session.
18. Open a terminal window on the booted system.
19. Log into the server as the root user.
20. Configure items in `/etc/sysconfig/hpe-auto-config`.
21. Run `hpe-auto-config` to configure the boot time options and other settings.

```
# hpe-auto-config
```

# OS installation using multiple ISO files

OS installation from ISO file on a PC requires using virtual media to simulate a local CD/DVD drive on the server. Some OSs require more than one DVD for a complete installation.

When using more than one DVD ISO, it is necessary to download a separate Java Application to mount the Virtual Media.

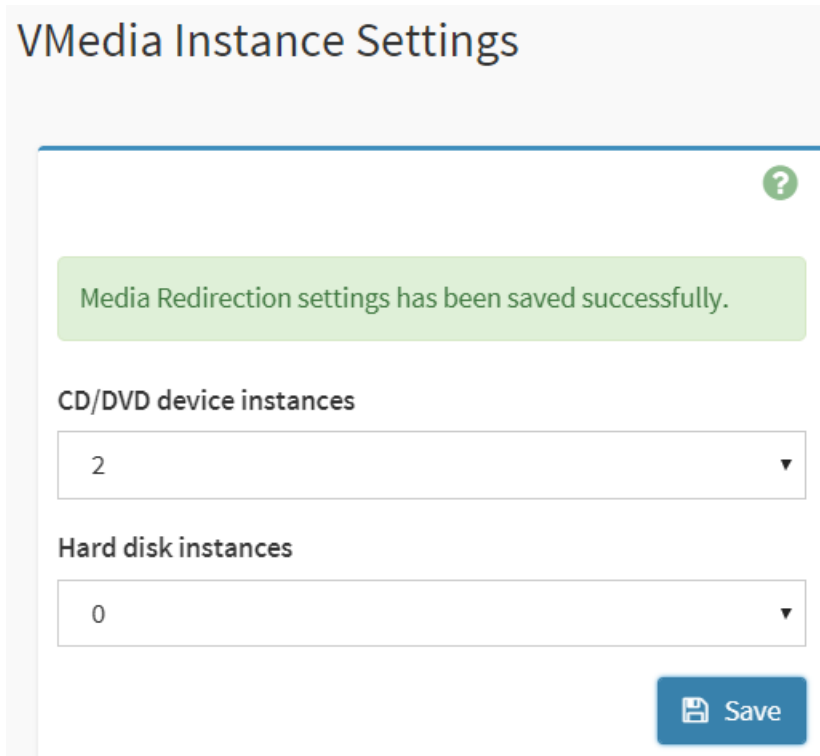
In this example, the SLES 15 OS is installed using two DVD ISO files.

## Prerequisites

- System administrator login has been completed.
- Launching JViewer from a macOS client is not supported.

## Procedure

1. On the home page, navigate to **Settings > Media Redirection > VMedia Instance Settings**.
2. Set the **CD/DVD device instances** to 2 (or more) and click **Save**.



3. Select **Remote Control** from the menu and click **Download Virtual Media Application**.



## Virtual Media Application

If mounting more than one virtual media device, or mounting a virtual Hard Disk, download and run this standalone application using 64-bit Java Runtime Environment.

Configure the Host IP setting to the IP address of the partition's monarch BMC.

 [Download Virtual Media Application](#)

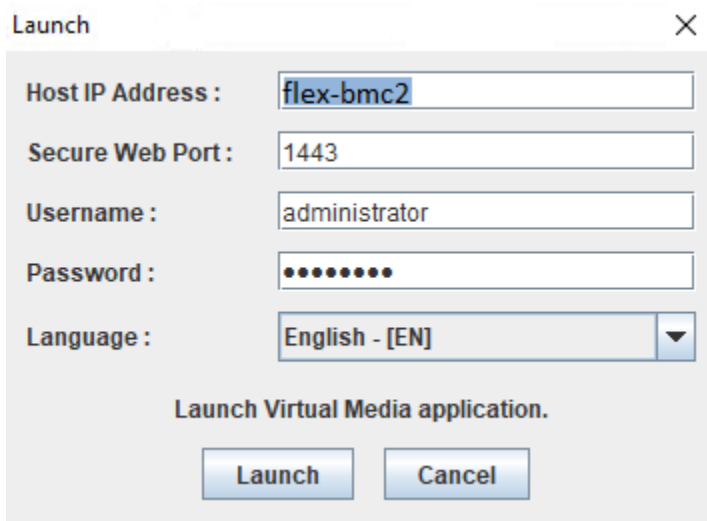
4. On your PC, open a command window ("cmd" or "PowerShell" in Windows) and navigate to the location where the `jviewer-sa.jar` file was downloaded. Run the `jviewer-sa.jar` file using the following syntax:

```
java -jar jviewer-sa.jar [-hostname <bmc_hostname>] [-u <username>] [-p <password>]
```

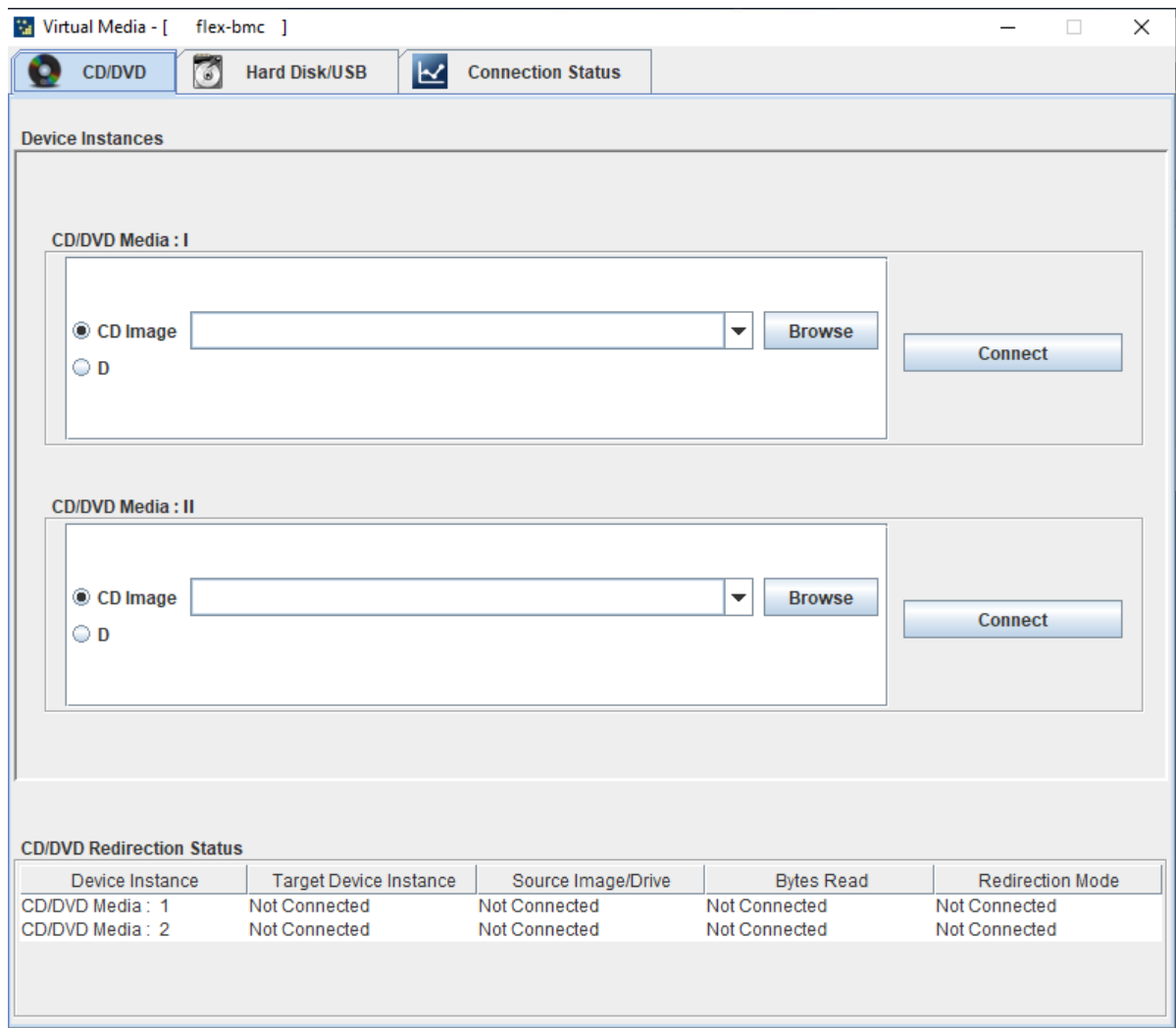
where `bmc_hostname` is the name of the monarch BMC in the partition.

**NOTE:** `java` must be in your PATH, otherwise specify the full path to your installed `java.exe` file.

5. Fill in the form with your system details and credentials and click **Launch**.



6. **Browse** to each of the DVD ISO files that are required for the OS installation and click **Connect**.



Once connected, JViewer will show information about each media device.

- Return to the RMC CLI and enter the following command to power On the system:

```
RMC cli> power on npar pnum=1
```

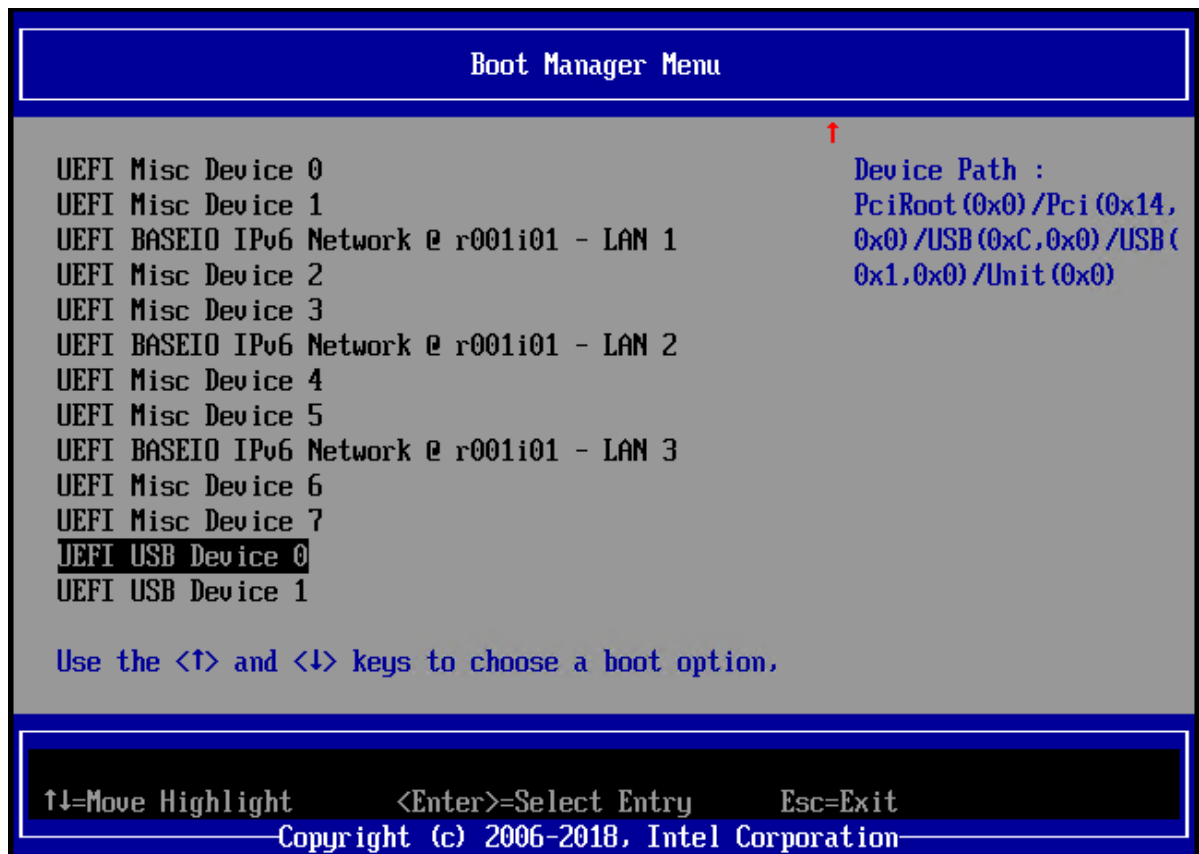
As the system powers up, the console will display the progress. You can view the console with the RMC text interface or the KVM interface. To view the system from the RMC text interface use:

```
RMC cli> connect npar pnum=p1
```

- When the system has booted to the UEFI Shell, exit to the Boot Options screen with the `exit` command and press **Enter**.

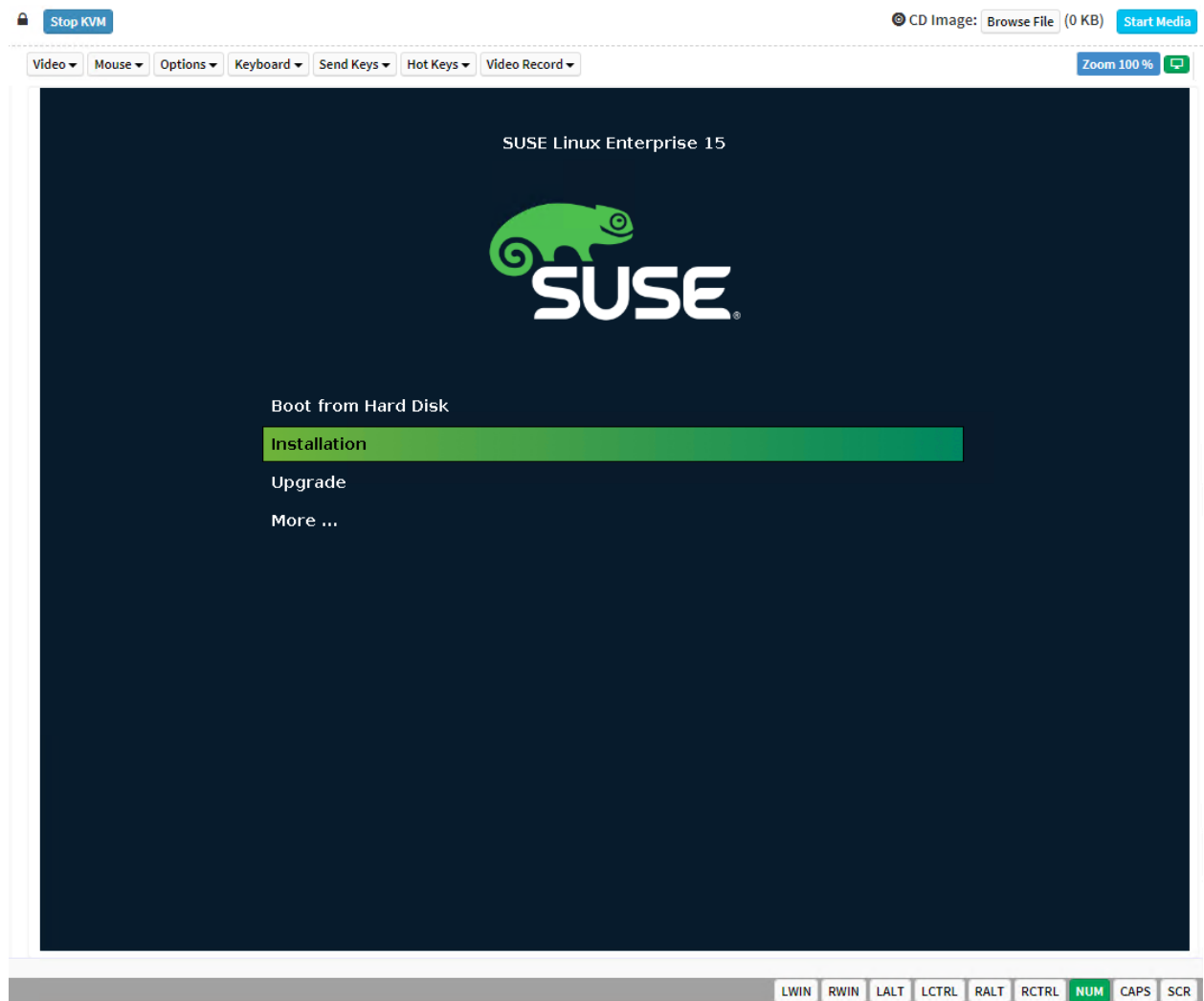
```
Shell> exit
```

- On the Boot Options screen, use the arrow keys to select **Boot Manager Menu** and press **Enter**.
- On the Boot Manager Menu screen, use the arrow keys to scroll through the list of devices and select **UEFI USB Device 0**. Press **Enter** to start booting from that device.



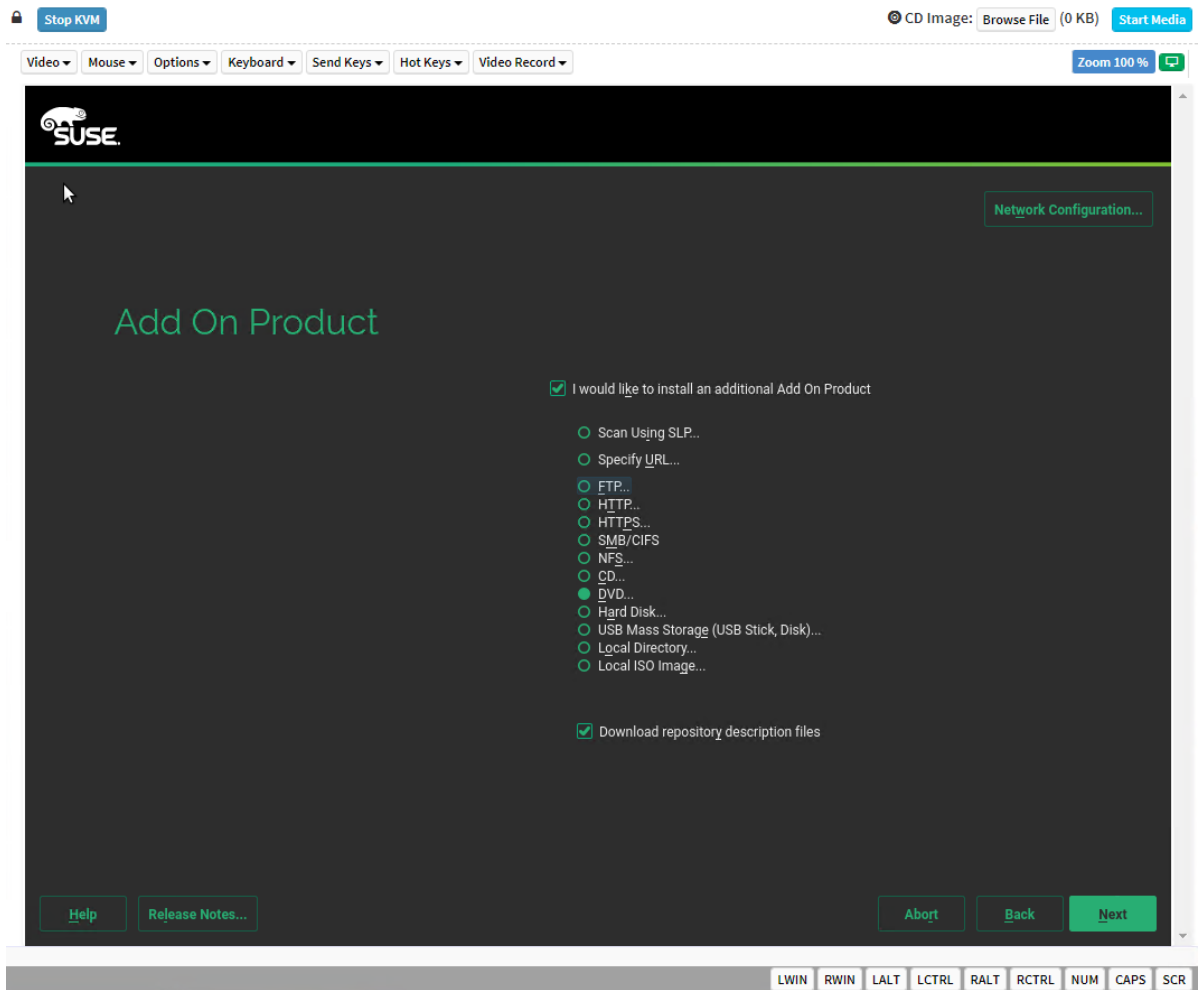
This virtual device named "UEFI USB Device 0", is the first "CD/DVD Image" that was mounted earlier. Notice that a second instance called "UEFI USB Device 1" is also displayed. This corresponds to the second DVD ISO file that was mounted earlier.

The OS starts to boot and presents the installation menu (SLES 15 in this example):

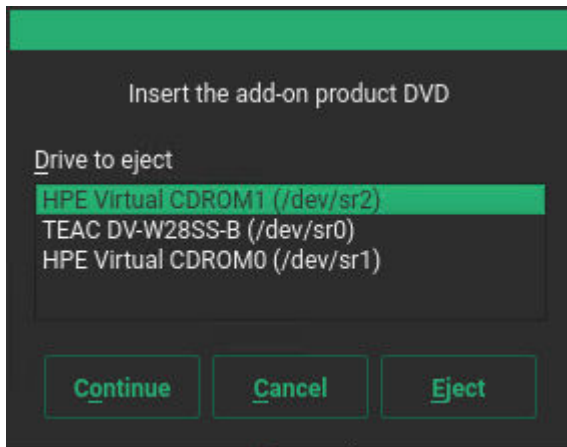


11. Follow the installation process until you reach the **Add on Product** screen. This will install content from the second Virtual CD/DVD. Select the following options:





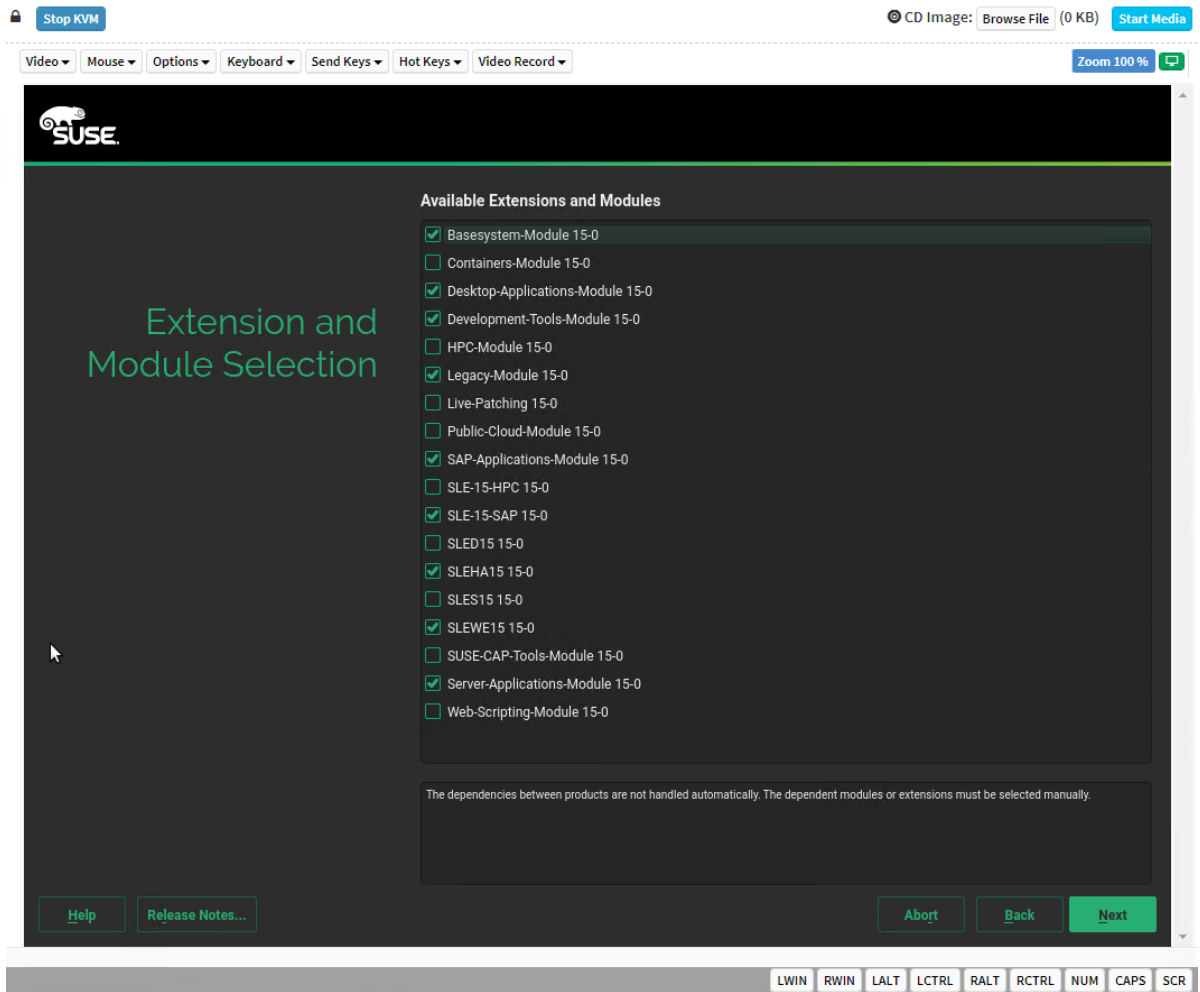
12. Select the HPE Virtual CDROM1 to install content from the second DVD ISO.



13. Select from the list of available extensions:







14. Continue with the OS installation. You can monitor the Virtual Media with the "Connection Status" tab.

| Virtual Media - [ h2-046-bmc2 ]                                                      |                        |                           |            |                  |
|--------------------------------------------------------------------------------------|------------------------|---------------------------|------------|------------------|
| <div> <div>CD/DVD</div> <div>Hard Disk/USB</div> <div>Connection Status</div> </div> |                        |                           |            |                  |
| CD/DVD Redirection Status                                                            |                        |                           |            |                  |
| Device Instance                                                                      | Target Device Instance | Source Image/Drive        | Bytes Read | Redirection Mode |
| CD/DVD Media : 1                                                                     | Virtual CD/DVD : 0     | C:\Users\User\Download... | 227630 KB  | Read Only        |
| CD/DVD Media : 2                                                                     | Virtual CD/DVD : 1     | C:\Users\User\Download... | 16121 KB   | Read Only        |

# Setting up boot order with the RMC web GUI

## Procedure

1. Log in to the RMC web GUI.
2. Click **nPartition** on the main screen or the menu bar on the left.
3. Click the **Boot Options** tab.
4. To verify the system boot order, check the **Boot Order** heading.
5. Change the boot order.
  - a. Click the  icon under the **Boot Order** heading.
  - b. Choose the boot option you want to change and click the pull-down list.
  - c. Select the position for the boot option from the list.
  - d. To confirm the changes, click **Submit**.
6. Reverify the system boot order. The system uses the listed boot order during the next system boot.



# Specifying boot options using the RMC CLI

The Superdome Flex 280 Server can be booted from multiple sources using RMC commands. The boot options can be specified during power on, power reset, or during reboot.

## Procedure

1. Log in to the RMC CLI.

2. Verify the boot option required.

The following boot options are available:

- *None* - No boot option specified. Boots from default source.
- *no\_oprom* - One time boot option that causes UEFI to boot without loading the default option ROMs.
- *BiosSetup* - Boot to BIOS setup.
- *Cd* - Boot from existing UEFI boot option entries that correspond to CD/DVD drives of any connection type (such as SATA and USB).
- *Hdd* - Boot from existing UEFI boot option entries that correspond to local hard disk drives, excluding USB drives.
- *Pxe* - Boot from existing UEFI boot option entries that correspond to PXE.
- *RemoteDrive* - Boot from existing UEFI boot option entries that correspond to remote (FibreChannel or iSCSI) hard disk drives.
- *SDCard* - Boot from existing UEFI boot option entries that correspond to SD cards.
- *Shell* or *UefiShell* - Boot to UEFI Shell.
- *UefiHttp* - Boot from existing UEFI boot option entries that correspond to HTTP boot.
- *Usb* - Boot from existing UEFI boot option entries that correspond to USB disk drives.

3. Verify if you are powering on the server, resetting the power, or rebooting the server.

- If you are powering on the server, enter the `power on` command.

```
power on npar pnum=0 [bootopt=BOOTOPT]
```

- If you are resetting the power or rebooting, enter the `power reset` or `reboot` command.

```
power reset npar pnum=0 [bootopt=BOOTOPT] [force]
```

```
reboot npar pnum=0 [bootopt=BOOTOPT] [force]
```

- If power is on, the `power reset` or `reboot` commands perform a graceful OS shutdown then restart the server.
- If `force` is specified, the commands perform an OS immediate (non-graceful) shutdown instead.



# Setting up boot order with UEFI

## Procedure

1. Interrupt the boot process and access UEFI.

Press **F2** to access the UEFI Boot Manager.

2. Select the **Boot Maintenance Manager** menu.

3. Select the **Change Boot Order** menu.

4. To select the boot options, press **Enter**.

5. Change the boot order.

- a. Use the up and down arrow keys to select a boot option.

- b. To move the boot option up, press **+**. To move the boot option down, press **-**.

- c. To finish setting the boot order, press **Esc**.

- d. To commit the changes and exit, press the down arrow and select **Commit Changes and Exit**. The changes take effect immediately and are applied on the next system boot.



# Secure boot

HPE Superdome Flex 280 Server systems support features that secure the boot process. When enabled, Secure Boot prevents execution of OS loaders, drivers, and UEFI applications that are not signed with an acceptable digital signature.

## Secure boot features

When secure boot is enabled on Superdome Flex 280 Server, system firmware verifies OS loader, driver, and UEFI application signatures before executing them.

By default, secure boot is disabled. This default applies to systems shipped from the factory.

Many secure boot configuration changes require resetting the system before booting an OS or accessing the UEFI Shell.

Secure boot protection applies both at the Boot Manager menu and at the UEFI Shell. In secure boot mode, the UEFI Shell disables the `mm` and `hexedit` commands.

System logs record changes to the secure boot mode. Secure boot checks performed during firmware verification also are logged.

## Default secure boot keys

The default keys include signatures for supported operating systems.

The Superdome Flex 280 Server default secure keys permit execution of images signed by the following certificates.

- HPE KEK 2016
- Microsoft Corporation KEK CA 2011
- SUSE Linux Enterprise Secure Boot CA
- HPE DB 2016
- HP DB 2013
- Microsoft Corporation UEFI CA 2011
- Microsoft Windows Production PCA 2011
- SUSE Linux Enterprise Secure Boot Signkey
- VMware certificate 2017

## Configuring Secure Boot on HPE Superdome Flex 280 Server

Secure Boot can be configured on Superdome Flex 280 Server through the RMC web GUI, the RMC CLI, or through UEFI.

### Procedure

- **Configure Secure Boot with the RMC web GUI**
- **Configure Secure Boot with the RMC CLI**
- **Configure Secure Boot with UEFI Boot Manager**



## Configuring Secure Boot with the RMC web GUI

### Procedure

1. Log in to the RMC web GUI.
2. Click **nPartition** from the main screen or the menu bar on the left.
3. Click the **Boot Options** tab.
4. To verify the status of Secure Boot, check the **Next Boot** entry under the **Secure Boot** heading.
5. Under the **Secure Boot** heading, toggle the **Next Boot** control to either enable or disable Secure Boot on the next system boot.
6. To apply the changes, reboot the system.

## Configuring Secure Boot with the RMC CLI

Secure Boot can be enabled and disabled with RMC CLI commands.

### Prerequisites

System must be powered off.

### Procedure

#### To enable Secure Boot:

1. Log in to the RMC CLI.
2. Verify if the system is powered off. If not, enter the `power off npar` command.
3. Enter the `modify npar` command.

- To enable Secure boot, enter the following command.

```
modify npar pnum=0 secure_boot=on
```

- To disable Secure boot, enter the following command.

```
modify npar pnum=0 secure_boot=off
```

4. Verify the Secure Boot state using the `show` command.

```
show npar verbose
```

5. Power on the system.

## Configuring Secure Boot with UEFI Boot Manager

### Procedure

1. Access UEFI Boot Manager from the nPartition console.
2. Access the **Secure Boot Configuration** menu.  
At Boot Manager, select the **Device Manager** menu, then select the **Secure Boot Configuration** menu.
3. Enable or disable secure boot.



To enable secure boot:

You can either enable with default keys, or install a custom set of keys.

- a. Enable Secure Boot with the default keys.

Select the **Attempt Secure Boot** option.

- b. Install custom keys.

Enable the **Custom Secure Boot Options** menu by changing the **Secure Boot Mode** setting to **Custom Mode**.

After installing custom keys, verify that the **Attempt Secure Boot** option is selected to enable secure boot.

To disable secure boot, clear the **Attempt Secure Boot** option.

4. To apply the changes, reset the system.

The system must be reset before you can load an OS or access the UEFI Shell. If you select "Continue" at the Boot Manager or attempt to use the Boot Manager to boot any option, a pop-up window will display:

Configuration changed. Reset to apply it now. Press ENTER to reset.

Press **Enter** to reset the system.

### More information

[Default secure boot keys](#)

## Installing or reinstalling default Secure Boot keys

When installing default keys, all secure boot data is written with a default set for supported operating systems.

### Prerequisites

The system automatically installs default keys if all Secure Boot keys have been deleted.

### Procedure

1. Access UEFI Boot Manager from the nPartition console.

2. Access the **Secure Boot Configuration** menu.

At Boot Manager select the **Device Manager** menu, then select the **Secure Boot Configuration** menu.

3. Select the **Custom Secure Boot Options** menu.

Change the **Secure Boot Mode** option to **Custom Mode**, then select the **Custom Secure Boot Options** menu.

4. Delete all KEK keys.

Select the **KEK Options** menu, then select the **Delete KEK** menu. For each key displayed, toggle the corresponding checkbox to delete the key.

5. Delete all DB keys.

Select the **DB Options** menu, then select the **Delete Signature** menu. For each key displayed, toggle the corresponding checkbox to delete the key.

6. Delete all DBX keys.

Select the **DBX Options** menu, then select the **Delete Signature** menu. Select the **Delete All Signature List** option and press **Y** to confirm.

7. Delete the PK key.



Select the **PK Options** menu, then select the **Delete Pk** checkbox. Press **Y** to confirm.

8. Reset the system to apply the changes.

You must reset the system before you can load an OS or access the UEFI Shell.

#### **More information**

[Default secure boot keys](#)





# Setting up remote media files with the RMC web GUI

HPE Superdome Flex 280 Server can access up to two ISO or image files on a remote file server.

## Prerequisites

- A file server configured on the local network using CIFS or NFS.
- Network address details for the file server.
- Credentials to access the file server.

## Procedure

1. Log in to the RMC web GUI.
2. Click **nPartition** from the main screen or the menu bar on the left.
3. Click the **Remote Console & Media** tab.

4. Under the **File Server** heading, click the configure icon (  ).

This enables you to configure the file server options, IP address, media path, and login credentials.

5. Connect to the file server by clicking **Connect**.
6. Select the image files.

Two image files can be selected and inserted.

To remove an image file, click **Eject**.

- a. Click the **Select Media Files** drop-down list under the **Media Files** heading.
- b. Select an image file in the specified file path.
- c. Click **Insert**. The selected image file is inserted.



# Websites

## HPE Superdome Flex 280 Server websites

- Product page  
[www.hpe.com/support/superdomeflex280-product](http://www.hpe.com/support/superdomeflex280-product)
- Customer documentation  
[www.hpe.com/support/superdomeflex280-docs](http://www.hpe.com/support/superdomeflex280-docs)
- Software  
[www.hpe.com/support/superdomeflex280-software](http://www.hpe.com/support/superdomeflex280-software)
- Hewlett Packard Enterprise server operating systems and virtualization software  
[www.hpe.com/us/en/servers/server-operating-systems.html](http://www.hpe.com/us/en/servers/server-operating-systems.html)
- HPE Superdome Flex 280 Server QuickSpecs  
[www.hpe.com/support/superdomeflex280-quickspecs](http://www.hpe.com/support/superdomeflex280-quickspecs)
- Spare parts list  
[www.hpe.com/support/superdomeflex280-spareparts](http://www.hpe.com/support/superdomeflex280-spareparts)
- Release sets (support matrix)  
[www.hpe.com/support/superdomeflex280-release-sets](http://www.hpe.com/support/superdomeflex280-release-sets)
- Safety and regulatory information  
[www.hpe.com/support/Safety-Compliance-EnterpriseProducts](http://www.hpe.com/support/Safety-Compliance-EnterpriseProducts)
- Recycling information  
[www.hpe.com/recycle](http://www.hpe.com/recycle)
- Visio templates  
[www.visiocal.com/hpe.htm](http://www.visiocal.com/hpe.htm)

The HPE-Integrity-MC stencil includes HPE Superdome Flex 280 Server front and rear physical shapes.

## HPE Superdome Flex 280 Server support documentation

HPE Superdome Flex 280 Server documentation for support specialists is available at [www.hpe.com/support/superdomeflex280-docs-restricted](http://www.hpe.com/support/superdomeflex280-docs-restricted) by signing in to [Hewlett Packard Enterprise Support Center](#) with an entitled account.

# Support and other resources

## Accessing Hewlett Packard Enterprise Support

- For live assistance, go to the Contact Hewlett Packard Enterprise Worldwide website:  
<https://www.hpe.com/info/assistance>
- To access documentation and support services, go to the Hewlett Packard Enterprise Support Center website:  
<https://www.hpe.com/support/hpesc>

### Information to collect

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

## Accessing updates

- Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.
- To download product updates:

### Hewlett Packard Enterprise Support Center

<https://www.hpe.com/support/hpesc>

### Hewlett Packard Enterprise Support Center: Software downloads

<https://www.hpe.com/support/downloads>

### My HPE Software Center

<https://www.hpe.com/software/hpesoftwarecenter>

- To subscribe to eNewsletters and alerts:  
<https://www.hpe.com/support/e-updates>
- To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:  
<https://www.hpe.com/support/AccessToSupportMaterials>





**IMPORTANT:** Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HPE Passport set up with relevant entitlements.

## Remote support

Remote support is available with supported devices as part of your warranty or contractual support agreement. It provides intelligent event diagnosis, and automatic, secure submission of hardware event notifications to Hewlett Packard Enterprise, which will initiate a fast and accurate resolution based on your product's service level. Hewlett Packard Enterprise strongly recommends that you register your device for remote support.

If your product includes additional remote support details, use search to locate that information.

### Remote support and Proactive Care information

#### HPE Get Connected

<https://www.hpe.com/services/getconnected>

#### HPE Proactive Care services

<https://www.hpe.com/services/proactivecare>

#### HPE Datacenter Care services

<https://www.hpe.com/services/datacentercare>

#### HPE Proactive Care service: Supported products list

<https://www.hpe.com/services/proactivecaresupportedproducts>

#### HPE Proactive Care advanced service: Supported products list

<https://www.hpe.com/services/proactivecareadvancedsupportedproducts>

### Proactive Care customer information

#### Proactive Care central

<https://www.hpe.com/services/proactivecarecentral>

#### Proactive Care service activation

<https://www.hpe.com/services/proactivecarecentralgetstarted>

## Warranty information

To view the warranty information for your product, see the links provided below:

#### HPE ProLiant and IA-32 Servers and Options

<https://www.hpe.com/support/ProLiantServers-Warranties>

#### HPE Enterprise and Cloudline Servers

<https://www.hpe.com/support/EnterpriseServers-Warranties>

#### HPE Storage Products

<https://www.hpe.com/support/Storage-Warranties>

#### HPE Networking Products

<https://www.hpe.com/support/Networking-Warranties>

## Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at the Hewlett Packard Enterprise Support Center:

<https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>



### **Additional regulatory information**

Hewlett Packard Enterprise is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements such as REACH (Regulation EC No 1907/2006 of the European Parliament and the Council). A chemical information report for this product can be found at:

**<https://www.hpe.com/info/reach>**

For Hewlett Packard Enterprise product environmental and safety information and compliance data, including RoHS and REACH, see:

**<https://www.hpe.com/info/ecodata>**

For Hewlett Packard Enterprise environmental information, including company programs, product recycling, and energy efficiency, see:

**<https://www.hpe.com/info/environment>**

## **Documentation feedback**

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